

Exercise 5-5: Get data from a JSON file

Did you use a language model?

A : No, I did not find any use for a language model for this assignment except for writing comments.

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In [2]: # Imports the Pandas Library
import pandas as pd
```

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In [3]: # Import the request module from urllib for URL operations like file downloads.
from urllib import request

# Define the URL of the JSON file containing shot data, intended for analysis or
shots_url = 'https://www.murach.com/python_analysis/shots.json'

# Use the urlretrieve function to download the JSON file from the specified URL,
request.urlretrieve(shots_url, filename='shots.json')
```

```
Out[3]: ('shots.json', <http.client.HTTPMessage at 0x251085280d0>)
```

```
In [4]: # Import the json module to parse JSON data and pandas for data manipulation.
import json
import pandas as pd

# Open and Load the JSON data from 'shots.json' into a Python dictionary.
with open('shots.json') as json_data:
    shots = json.load(json_data)

# Extract the 'rowSet' data, which contains the actual shot data, from the first
allRows = shots['resultSets'][0]['rowSet']

# Extract and convert the 'headers' to lowercase, which will be used as column names
columnHeaders = [x.lower() for x in shots['resultSets'][0]['headers']]

# Create a pandas DataFrame 'shots' with the shot data and the column names set
shots = pd.DataFrame(data=allRows, columns=columnHeaders)

# Display the first five rows of the DataFrame to verify the structure and content
shots.head()
```

Out[4]:

	grid_type	game_id	game_event_id	player_id	player_name	team_id	team_na
0	Shot Chart Detail	0020900015	4	201939	Stephen Curry	1610612744	Golc St Warri
1	Shot Chart Detail	0020900015	17	201939	Stephen Curry	1610612744	Golc St Warri
2	Shot Chart Detail	0020900015	53	201939	Stephen Curry	1610612744	Golc St Warri
3	Shot Chart Detail	0020900015	141	201939	Stephen Curry	1610612744	Golc St Warri
4	Shot Chart Detail	0020900015	249	201939	Stephen Curry	1610612744	Golc St Warri

5 rows × 24 columns

In []: